

Презентация по лабораторной работе №2

Задача о погоне

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Информация

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Выполнение работы

```
using Plots using DifferentialEquations
const a = 6.4 const n = 2.4
const r0 = a/(n+1) const r0_2 = a/(n-1)
const T = (0, 2*pi) const T_2 = (-pi, pi)
function F(u, p, t) return u / sqrt(n*n-1) end
problem = ODEProblem(F, r0, T)
result = solve(problem, abstol=1e-8, reltol=1e-8) [show?] result.u [show?] result.t
dxR = rand(1:size(result.t)[1]) rAngels = [result.t[dxR] for i in 1:size(result.t)[1]]
plt = plot(proj=:polar, aspect_ratio=:equal, dpi = 1000, legend = true, bg=:white)
plot!(plt, xlabel = "theta", ylabel = "r(t)", title="Случай №1", legend=:outerbottom)
plot!(plt, [rAngels[1], rAngels[2]], [0.0, result.u[size(result.u)[1]]], label = "Путь лодки", color=:blue, lw=1)
plot!(plt, result.t, result.u, xlabel="theta", ylabel="r(t)", label = "Путь катера", color=:green, lw=1)
scatter!($
savefig(plt, "lab2_01.png")

problem = ODEProblem(F, r0_2, T_2)
result = solve(problem, abstol = 1e-8, reltol=1e-8)
dxR = rand(1:size(result.t)[1])
rAngels = [result.t[dxR] for i 1:size(result.t)[1]]
plt1 = plot(proj=:polar, aspect_ratio=:equal, dpi = 1000, legend=true, bg=:white)
plot!(plt1, xlabel="theta", ylabel="r(t)", title="Случай № 2", legend=:outerbottom)
plot!(plt1, [rAngels[1], rAngels[2]], [0.0, result.u[size(result.u)[1]]], label = "Путь лодки", color=:blue, lw =1)
plot!(plt1, result.t, result.u, xlabel="theta", ylabel="r(t)", label = "Путь катера", color=:green, lw=1)
scatter!($
savefig(plt1, "lab2_02.png")
```

Рис. 1: Решение

```
→ work julia lab2.jl
  Resolving package versions...
No Changes to `~/.julia/environments/v1.11/Project.toml`
No Changes to `~/.julia/environments/v1.11/Manifest.toml`
  Resolving package versions...
No Changes to `~/.julia/environments/v1.11/Project.toml`
No Changes to `~/.julia/environments/v1.11/Manifest.toml`
→ work ls
Plots      cloned_hello hello      hello.git  lab2.jl    lab2_01.png lab2_02.png study
→ work
```

Рис. 2: Выполнение кода

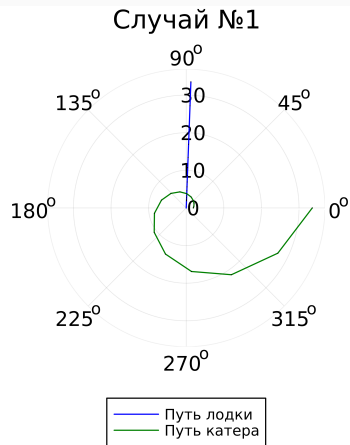


Рис. 3: График 1

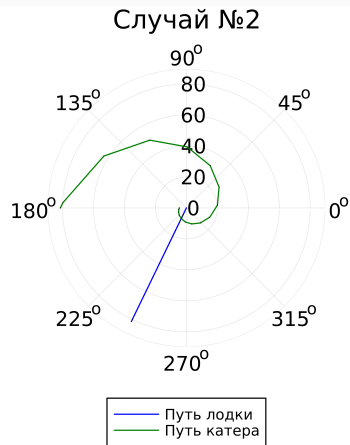


Рис. 4: График 2